

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Ans: Step 1: Number of segments

$$\frac{150,000,000}{1500} = 100,000$$

Segments = 100,000

Step 2: Number of frames

Each segment is placed into one frame.

Frames = 100,000

Step 3: Total Data Link overhead

$$100,000 \times 40 = 4,000,000 \text{ bytes}$$

Overhead = 4 MB

Step 4: Total data transmitted

$$150 + 4 = 154 \text{ MB}$$

Convert to bits:

$$154 \times 8 = 1232 \text{ Mbits}$$

Transmission time

$$\frac{1232}{50} = 24.64 \text{ s}$$

Answer

- Segments = 100,000
- Frames = 100,000
- Overhead = 4 MB
- Transmission time = 24.64 s

Reliability

- Transport Layer: segmentation, acknowledgements, error recovery, flow control.
- Data Link Layer: framing, error detection (CRC), retransmission of damaged frames.

2. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given

- Window size = 6
- Total frames = 48

Windows required

$$\frac{48}{6} = 8$$

Windows = 8

Retransmissions

1 frame lost in every window.

$$8 \times 1 = 8$$

Retransmissions = 8

Flow Control

Flow control prevents the sender from overwhelming the receiver, reducing congestion and improving efficiency.

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20 byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given

- Packet = 12,000 bytes
- MTU = 1500 bytes
- IP Header = 20 bytes

Payload per fragment

$$1500 - 20 = 1480 \text{ bytes}$$

Number of fragments

$$\left\lceil \frac{12000}{1480} \right\rceil = 9$$

Fragments = 9

Header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Answer

- Fragments = 9
- Header overhead = 180 bytes

Network Layer

Uses fragment offset, identification, and MF flag so the destination can correctly reassemble all fragments.

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every

window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Sliding window size = 6 frames
- Total frames = 48
- Each frame = 1024 bytes

Step 1: Number of transmission windows

$$\frac{48}{6} = 8$$

Transmission windows = 8

Step 2: Total retransmissions

One frame is lost in each window.

$$8 \times 1 = 8$$

Retransmissions = 8

Flow Control

Flow control ensures that the sender does not send data faster than the receiver can process it. This prevents buffer overflow, reduces packet loss, minimizes retransmissions, and improves communication efficiency.

Final Answer

- Transmission windows required: 8
- Total retransmissions: 8
- Flow control: Prevents sender overload, reduces congestion, and improves reliable communication.

5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the

amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given

- File = 600 MB
- Checkpoint every 60 MB
- Failure after 390 MB

Checkpoints before failure

Completed checkpoints:

60,120,180,240,300,360

Total checkpoints = 6

Retransmitted data

$$390 - 360 = 30 \text{ MB}$$

Retransmitted = 30 MB

Importance

Synchronization checkpoints allow communication to resume from the last checkpoint instead of restarting the entire transfer.

6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given

- Original = 900 MB
- Compression = 40%
- Encryption overhead = 5%

Compressed size

$$900 \times 0.60 = 540 \text{ MB}$$

Encrypted size

$$540 \times 1.05 = 567 \text{ MB}$$

Reduction

$$\frac{900 - 567}{900} \times 100 = 37\%$$

Answer

- Compressed = 540 MB
- Encrypted = 567 MB
- Reduction = 37%

Presentation Layer

Provides compression for faster transmission and encryption for secure communication.

7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given

- Data = 3 GB = 3000 MB
- Throughput = 120 Mbps
- Request size = 3 MB

Requests

$$\frac{3000}{3} = 1000$$

Convert data to bits

$$3000 \times 8 = 24000 \text{ Mbits}$$

Transmission time

$$\frac{24000}{120} = 200 \text{ s}$$

Answer

- Requests = 1000
- Transmission time = 200 seconds

Application Layer

Provides services like FTP, user authentication, file transfer, and error handling.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Processing delay

Application = 4 ms

Transport = 3 ms

Network = 5 ms

Data Link = 2 ms

Physical = 1 ms

Total processing:

$$4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

Total delay

$$15 + 18 + 12 = 45 \text{ ms}$$

Answer

Total delay = 45 ms

Layer Contributions

- Application: user services
- Transport: segmentation & reliability
- Network: routing
- Data Link: framing & error detection
- Physical: signal transmission

9. A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Overhead = 80 bytes

Useful payload/frame

$$1600 - 80 = 1520 \text{ bytes}$$

Frames

$$\left\lceil \frac{80,000,000}{1520} \right\rceil = 52,632$$

Total overhead

$$52,632 \times 80 = 4,210,560 \text{ bytes}$$

$$\approx 4.21 \text{ MB}$$

Efficiency

$$\frac{1520}{1600} \times 100 = 95\%$$

Answer

- Frames = 52,632
- Overhead = 4,210,560 bytes (≈ 4.21 MB)
- Efficiency = 95%

Protocol Overhead

Higher overhead reduces usable bandwidth and lowers transmission efficiency.

10. A hospital transfers a patient's medical record of 240 MB through an IoT enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given

- Original = 240 MB
- Compression = 30%
- Segment = 1400 bytes
- Frame overhead = 60 bytes

Compressed size

$$240 \times 0.70 = 168 \text{ MB}$$

Compressed bytes

168,000,000

Segments

$$\frac{168,000,000}{1400} = 120,000$$

Protocol overhead

$$120,000 \times 60 = 7,200,000 \text{ bytes}$$

Final transmitted data

$$168,000,000 + 7,200,000 = 175,200,000 \text{ bytes}$$

$$= 175.2 \text{ MB}$$

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand